Farah F. Cisse

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EDUCATION

University of California, Berkeley – College of Computing & Data Science; College of Engineering Berkeley, CA
B.A. in Data Science and B.S. in Bioengineering – GPA: 3.7 Graduating Dec 2026

Honors: STEM Excellence through Equity & Diversity (SEED) Honor Scholar, Bioengineering Scholar, Bioengineering Honors Society

Relevant Coursework: Machine Learning, Data Science, Data Analytics, Data Structures & Algorithms, Probability & Statistics, Computational Molecular & Cell Biology, Computational Functional Genomics, Linear Algebra & Differential Equations

SKILLS & INTERESTS

Skills: Python (Pandas, Matplotlib, NumPy, SciPy), Machine Learning (statsmodels, Scikit-Learn), MATLAB, Postgres, SQL, Java, R, Data Science (Manipulation/Cleaning, A/B Testing), Prompt engineering, Feature engineering, IDE (Jupyter Notebook, IntelliJ, RStudio, VS-code), Bioinformatics (Jbrowse, AlphaFold, Chimera, BLAST, FoldSeek, Cytoscape), Github, Bash, Streamlit, Supabase

Interests: AI/ML-integrated automation; data science for healthcare & environmental sustainability; bioinformatics, biotech, food tech

Hobbies: calisthenics (2nd place in a 2025 USCCO competition), volleyball, fitness, hiking, traveling, mentoring, reading

WORK EXPERIENCE

Marine Biologics (a biotech startup, AI-powered platform transforming seaweed into clean-label ingredients) Berkeley, CA
*Data Intern – **Project:** Building a Data Ingestion Pipeline* Jun 2025 – Aug 2025
- Automated PDF-to-CSV conversion of scientific publication data using open-source tools (Docling, Crossref, GNU Parallel, LLMs), enabling large-scale data collection for the company database, enhancing ML-assisted development of marine-derived ingredients
- Built & launched a web app for seamless PDF upload & real-time data visualization via an interactive dashboard, saving cost & time

Staller Lab | UC Berkeley Center for Computational Biology Berkeley, CA
*Computational Biology Researcher – **Project:** Repression Domains (RD) in Autism Spectrum Disorder (ASD)* Sep 2024 – Jan 2025
- Compiled and cleaned (using Python) a 3k+ repression domain annotation dataset from DelRosso, Tycko, and Soto Research Papers
- Discovered the proportion of autism-associated transcription factors that contain RD by using an enrichment analysis (BEDTools, Fisher's exact tests), revealing these domains as intermediate-frequency sites for ASD patient variants and potential therapeutic targets

Lawrence Berkeley National Laboratory | Algal & Fungal Genomics Berkeley, CA
*Bioinformatics Researcher – **Project:** Predicting Unknown Protein Functions* in *Chlamydomonas reinhardtii* May 2024 – Jan 2025
- Explored unknown proteins in similarity networks (using Cytoscape), identifying trends by overlaying pathway & fold-change data
- Created Bash scripts to run ProteinCartography, identifying similar known proteins to strengthen annotations of unknown proteins
- Generated insights on protein structure & sequence similarities from FoldSeek & BLAST results, improving protein annotations

Control for Assistive & REhabilitation and Robotics (CARE) Lab | San Francisco State University San Francisco, CA
*Biomechanics Researcher – **Project:** Developing Smart Shoe Insole Sensor for Gait Analysis* Jun 2023 – Aug 2023
- Obtained & analyzed real-time gait parameters (using MATLAB, Arduino, Eagle, Excel) to potentially detect gait abnormalities early
- Validated the accuracy of the shoe insole sensor against thigh angle data, an industry-recognized but more complex & costly method
- Developed an experimental protocol for human subject testing of the shoe insole sensor for future research cohorts

PROJECT EXPERIENCE

Effects of General Anesthesia (GA) on Immature and Mature Human Brains | Public Health & Health Science ULAB
- Co-author of a literature review on GA impacts on neurocognition and brain activity (Research Publication)

Unsupervised Clustering and Differential Expression Analysis of Lung scRNA-seq Data (~23k cells × 1.9k genes)
- Author of an unsupervised machine learning analysis using Principal Component Analysis (PCA), K-means clustering, differential expression, and annotation-based validation to identify biologically meaningful lung cell populations (Project Paper)

LEADERSHIP EXPERIENCE

UC Berkeley Calisthenics Club | *Executive Officer* Aug 2025 – Present
- Lead skill stations and teach fundamental techniques to 20+ college students; manage social media, increasing followers by 40%

UC Berkeley College of Computing, Data Science and Society | *Data Science Tutor* Aug 2024 – Jan 2025
- Taught Python and Data Science to 70+ college students in weekly tutoring sections and office hours

UC Berkeley Bioengineering Honors Society | *Senior Advisor, Secretary, and Outreach Chair* Jan 2023 – Dec 2024
- Led a team of 7 to organize 12 outreach events introducing bioengineering to 200+ elementary through college students

UC Berkeley Society of Women Engineers | *STEM Outreach Chair* Sep 2022 – Jan 2024
- Led a team of 8 in organizing monthly STEM events for 80+ elementary and middle school students

Other affiliations: TalkShop mentorship and content creation, RePicture STEM career accelerator, Cal volleyball intramurals